

21 Compress and decompress files using tar.

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ mkdir sample
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cd sample/
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ touch a b c d e text
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ tar cvf txt.tar text
text
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ ls
a b c d e text txt.tar
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ rm text
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ ls
a b c d e txt.tar
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ tar xvf txt.tar
text
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ ls
a b c d e text txt.tar
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$
```

1. Purpose

Demonstrate basic file archiving and extraction using tar.

2. Commands Executed

- Created sample directory and files.
- Created txt.tar using tar -cvf.
- Deleted text using rm.
- Restored text using tar -xvf.

3. Verification

Verified file creation and restoration using ls.

4. Conclusion

Successfully created, extracted, and restored a file using tar.

22 . Monitor system resources with top or htop.

```
top - 10:53:34 up 39 min, 1 user, load average: 0.01, 5.45, 11.34
Tasks: 175 total, 1 running, 174 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.4 us, 1.2 sy, 0.0 ni, 98.0 id, 0.0 wa, 0.0 hi, 0.4 si, 0.0 st
MiB Mem : 1968.5 total, 82.8 free, 750.6 used, 1135.1 buff/cache
MiB Swap: 2680.0 total, 2679.5 free, 0.5 used, 1028.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 1290 vamshik+ 20   0 3771892 345896 136412 S   3.0  17.2   0:32.75 gnome-shell
    28 root       20   0        0        0        0 S   0.3   0.0   0:00.14 kcompactd0
    58 root       20   0        0        0        0 I   0.3   0.0   0:01.98 kworker/u2:5-flush-8:0
 1745 vamshik+ 20   0 555740 54916 41764 S   0.3   2.7   0:03.07 gnome-terminal-
2998 vamshik+ 20   0 13100 4168 3364 R   0.3   0.2   0:00.07 top
    1 root       20   0 167900 13208 8388 S   0.0   0.7   0:02.54 systemd
    2 root       20   0        0        0        0 S   0.0   0.0   0:00.00 kthreadd
    3 root       20   0        0        0        0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root       0 -20        0        0        0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
    5 root       0 -20        0        0        0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
    6 root       0 -20        0        0        0 I   0.0   0.0   0:00.00 kworker/R-slub_
    7 root       0 -20        0        0        0 I   0.0   0.0   0:00.00 kworker/R-netns
   11 root       20   0        0        0        0 I   0.0   0.0   0:00.94 kworker/u2:0-events_unbound
   12 root       0 -20        0        0        0 I   0.0   0.0   0:00.00 kworker/R-mm_pe
   13 root       20   0        0        0        0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   14 root       20   0        0        0        0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   15 root       20   0        0        0        0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   16 root       20   0        0        0        0 S   0.0   0.0   0:01.51 ksoftirqd/0
   17 root       20   0        0        0        0 I   0.0   0.0   0:00.42 rcu_preempt
   18 root       rt    0        0        0        0 S   0.0   0.0   0:00.03 migration/0
   19 root      -51   0        0        0        0 S   0.0   0.0   0:00.00 idle_inject/0
   20 root       20   0        0        0        0 S   0.0   0.0   0:00.00 cpuhp/0
   21 root       20   0        0        0        0 S   0.0   0.0   0:00.00 kdevtmpfs
   22 root       0 -20        0        0        0 I   0.0   0.0   0:00.00 kworker/R-inet_
   23 root       20   0        0        0        0 S   0.0   0.0   0:00.00 kauditd
```

1. Purpose

Monitor CPU, memory, load average, and running processes using top.

2. Command Execution & Verification

Used top to view real-time system and process information.

3. Conclusion

Verified system performance and running processes using top

23 .Create and manage user groups.

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo groupadd developers
[sudo] password for vamshikumar:
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,vamshikumar
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:vamshikumar
floppy:x:25:
tape:x:26:
sudo:x:27:vamshikumar
audio:x:29:pulse
dip:x:30:vamshikumar
www-data:x:33:
backup:x:34:
operator:x:37:
list:x:38:
irc:x:39:
src:x:40:
nats:x:41:

vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo useradd techie
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo passwd techie
New password:
Retype new password:
passwd: password updated successfully
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo usermod -aG developers techie
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ groups techie
techie : techie developers
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo gpasswd -d techie developers
Removing user techie from group developers
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo groupdel developers
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ groups techie
techie : techie
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ sudo userdel techie
vamshikumar@vamshikumar-VirtualBox:~/Desktop/sample$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105:/:nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111:/:home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/:nonexistent:/usr/sbin/nologin
tss:x:106:113:TPM software stack,,,:/var/lib/tpm:/bin/false
```

```
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
rtkit:x:116:123:RealtimeKit,,,:/proc:/usr/sbin/nologin
whoopsie:x:117:124::/nonexistent:/bin/false
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
fwupd-refresh:x:120:126:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
nm-openvpn:x:121:127:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saned:x:122:129::/var/lib/saned:/usr/sbin/nologin
colord:x:123:130:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:124:131::/var/lib/geoclue:/usr/sbin/nologin
pulse:x:125:132:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:126:65534::/run/gnome-initial-setup:/bin/false
hplip:x:127:7:HPLIP system user,,,:/run/hplip:/bin/false
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
vamshikumar:x:1000:1000:vamshikumar,,,:/home/vamshikumar:/bin/bash
```

1. Purpose

Create and manage Linux user groups.

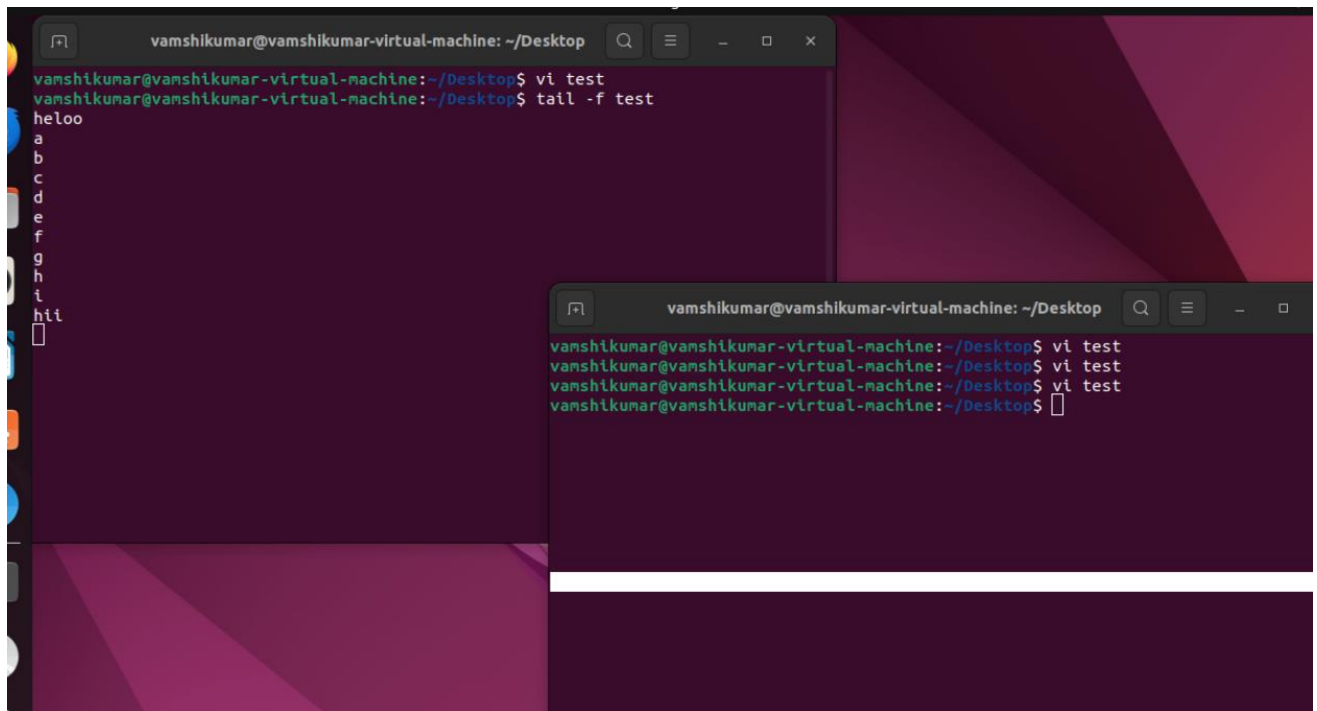
2. Command Execution & Verification

Created developers, added techie, verified group membership, and removed the group.

3. Conclusion

Successfully practiced user and group management commands.

- 24 .Set up SSH password less authentication.
25. Monitor log files using tail or grep.



```
vamshikumar@vamshikumar-virtual-machine: ~/Desktop
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ vi test
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ tail -f test
heloo
a
b
c
d
e
f
g
h
i
t
hi
[]

vamshikumar@vamshikumar-virtual-machine: ~/Desktop
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ vi test
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ vi test
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ []
```

26. Set up a web server (e.g., Apache or Nginx).

Installed Apache2, started the service, and verified its status using systemctl.

3. Browser Verification

Verified the Apache2 default page successfully loaded in the browser using localhost.

4. Conclusion

Successfully installed, started, and verified the Apache2 web server.

27 Configure and secure a MySQL Database

```
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libaio1 libcgi-fast-perl libcgi-pm-perl libevent-core-2.1-7 libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-template-perl libmecab2 libprotobuf-lite23 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libipc-sharedcache-perl mailx tinycd
The following NEW packages will be installed:
  libaio1 libcgi-fast-perl libcgi-pm-perl libevent-core-2.1-7 libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-template-perl libmecab2 libprotobuf-lite23 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0
0 upgraded, 20 newly installed, 0 to remove and 450 not upgraded.
Need to get 29.6 MB of archives.
After this operation, 243 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 mysql-common all 5.8+1.0.8 [7,212 B]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-client-core-8.0 amd64 8.0.46-0ubuntu0.22.04.3 [2,742 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-client-8.0 amd64 8.0.46-0ubuntu0.22.04.3 [22.7 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libaio1 amd64 0.3.112-13build1 [7,176 B]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libevent-core-2.1-7 amd64 2.1.12-stable-1build3 [93.9 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libevent-pthreads-2.1-7 amd64 2.1.12-stable-1build3 [7,642 B]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libmecab2 amd64 0.996-14build9 [199 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libprotobuf-lite23 amd64 3.12.4-1ubuntu7.22.04.6 [209 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-server-core-8.0 amd64 8.0.46-0ubuntu0.22.04.3 [17.8 MB]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-server-8.0 amd64 8.0.46-0ubuntu0.22.04.3 [1,451 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libcgi-pm-perl all 4.54-1 [188 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libfcgi0ldbl amd64 2.4.2-2ubuntu0.1 [28.1 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libfcgi-perl amd64 0.82+ds-1build1 [22.8 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libcgi-fast-perl all 1:2.15-1 [10.5 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libfcgi-bin amd64 2.4.2-2ubuntu0.1 [11.1 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libhtml-template-perl all 2.97-1.1 [59.1 kB]
Get:17 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 mecab-utils amd64 0.996-14build9 [4,850 B]
Get:18 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 mecab-ipadic all 2.7.0-20070801+main-3 [6,718 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 mecab-ipadic-utf8 all 2.7.0-20070801+main-3 [4,384 B]
Get:20 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-server all 8.0.46-0ubuntu0.22.04.3 [9,462 B]
Fetched 29.6 MB in 15s (2,030 kB/s)
Preconfiguring packages ...
Selecting previously unselected package mysql-common.
```

```
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-10 11:14:57 IST; 28s ago
     Process: 3971 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 3979 (mysqld)
      Status: "Server is operational"
        Tasks: 38 (limit: 2268)
       Memory: 369.6M
          CPU: 1.965s
       CGroup: /system.slice/mysql.service
               └─3979 /usr/sbin/mysqld

Aug 10 11:14:55 vamshikumar-VirtualBox systemd[1]: Starting MySQL Community Server...
Aug 10 11:14:57 vamshikumar-VirtualBox systemd[1]: Started MySQL Community Server.
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sudo systemctl enable mysql
Synchronizing state of mysql.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable mysql
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46-0ubuntu0.22.04.3 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> yes
->
[2]+  Stopped                  sudo mysql
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.
```

```

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: y

There are three levels of password validation policy:

LOW      Length >= 8
MEDIUM  Length >= 8, numeric, mixed case, and special characters
STRONG Length >= 8, numeric, mixed case, special characters and dictionary file

Please enter 0 = LOW, 1 = MEDIUM and 2 = STRONG: 2

Skipping password set for root as authentication with auth_socket is used by default.
If you would like to use password authentication instead, this can be done with the "ALTER_USER" command.
See https://dev.mysql.com/doc/refman/8.0/en/alter-user.html#alter-user-password-management for more information.

By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
a user account created for them. This is intended only for
testing, and to make the installation go a bit smoother.
You should remove them before moving into a production
environment.

Remove anonymous users? (Press y|Y for Yes, any other key for No) : y
Success.

Normally, root should only be allowed to connect from
'localhost'. This ensures that someone cannot guess at
the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : y
Success.

By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
a user account created for them. This is intended only for
testing, and to make the installation go a bit smoother.
You should remove them before moving into a production
environment.

Remove anonymous users? (Press y|Y for Yes, any other key for No) : y
Success.

Normally, root should only be allowed to connect from
'localhost'. This ensures that someone cannot guess at
the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : y
Success.

By default, MySQL comes with a database named 'test' that
anyone can access. This is also intended only for testing,
and should be removed before moving into a production
environment.

Remove test database and access to it? (Press y|Y for Yes, any other key for No) : y
- Dropping test database...
Success.

- Removing privileges on test database...
Success.

Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : y
Success.

All done!
vamshikumar@vamshikumar-VirtualBox:~/Desktop$

```

Purpose

Setting up a web server like Apache or Nginx enables hosting web pages and applications by servicing incoming HTTP/HTTPS requests on specific network ports.

Command Execution and Verification

Running `sudo systemctl status apache2` shows a failed state because port 80 is already in use (Address already in use: AH00072: make_sock: could not bind to address 0.0.0.0:80). In contrast, running `sudo systemctl status nginx` shows active (running) with a master process and worker processes, confirming Nginx is currently handling port 80

successfully.

Conclusion

Only one web server can bind to port 80 at a time, so starting Apache while Nginx is already active causes a port collision error until one service is stopped or reconfigured to a different port.

28 .Set up a Application Server (e.g.,Apache Tomcat)

The screenshot shows the process of installing and setting up Apache Tomcat 10.1.57 on a Linux system. The terminal window displays the following commands and output:

```
vamshikumar@vamshikumar-VirtualBox:/opt$ sudo wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.46/bin/apache-tomcat-10.1.46.tar.gz
--2026-08-10 15:21:14-- https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.46/bin/apache-tomcat-10.1.46.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-08-10 15:21:15 ERROR 404: Not Found.

vamshikumar@vamshikumar-VirtualBox:/opt$ ls /opt/tomcat
bin BUILDING.txt conf CONTRIBUTING.md lib LICENSE logs NOTICE README.md RELEASE-NOTES RUNNING.txt temp webapps work
vamshikumar@vamshikumar-VirtualBox:/opt$ ll
total 14076
drwxr-xr-x 3 root root 4096 Aug 10 15:26 ./
drwxr-xr-x 20 root root 4096 Aug 4 06:22 ../
-rw-r--r-- 1 root root 14400273 Jul 3 18:43 apache-tomcat-10.1.57.tar.gz
drwxr-xr-x 9 root root 4096 Aug 10 15:25 tomcat/
vamshikumar@vamshikumar-VirtualBox:/opt$ sudo chmod +x /opt/tomcat/bin/*.ssh
chmod: cannot access '/opt/tomcat/bin/*.ssh': No such file or directory
vamshikumar@vamshikumar-VirtualBox:/opt$ sudo chmod +x /opt/tomcat/bin/*.sh
chmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
vamshikumar@vamshikumar-VirtualBox:/opt$
```

The web browser window shows the Apache Tomcat 10.1.57 homepage. The page includes a navigation bar with links to Home, Documentation, Configuration, Examples, Wiki, and Mailing Lists. The main content area features a green banner with the message "If you're seeing this, you've successfully installed Tomcat. Congratulations!" and a recommended reading section with links to Security Considerations, Manager Application, and Clustering/Session Replication. Below this is a Developer Quick Start section with links to Tomcat Setup, Realms & AAA, Examples, and Servlet Specifications. The footer contains three columns of links: Managing Tomcat, Documentation, and Getting Help.

Purpose

Setting up an application server like Apache Tomcat allows Java-based web applications and servlets to run and serve dynamic content over HTTP.

Command Execution and Verification

The command `sudo systemctl status tomcat9` checks the status of the Tomcat service, returning active (running) with process logs showing initializing ProtocolHandler ["http-nio-8080"] and Server startup in [2251] milliseconds. This verifies that Apache Tomcat is successfully initialized and running on port 8080.

Conclusion

Deploying an application server requires verifying the service state via `systemctl` to confirm that the Java runtime environment and web application directories are deployed and listening on the designated port.

29 ,create a service file for Apache Tomcat.(Should execute by using `systemctl` command)

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo systemctl start apache2
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-10 11:24:33 IST; 16s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 5011 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 5015 (apache2)
       Tasks: 55 (limit: 2268)
      Memory: 5.1M
         CPU: 50ms
    CGroup: /system.slice/apache2.service
            └─5015 /usr/sbin/apache2 -k start
              └─5016 /usr/sbin/apache2 -k start
                └─5017 /usr/sbin/apache2 -k start

Aug 10 11:24:33 vamshikumar-VirtualBox systemd[1]: Starting The Apache HTTP Server...
Aug 10 11:24:33 vamshikumar-VirtualBox apachectl[5014]: AH00558: apache2: Could not reliably determine the server's fully qualified
Aug 10 11:24:33 vamshikumar-VirtualBox systemd[1]: Started The Apache HTTP Server.
lines 1-17/17 (END)
```

Purpose

Setting up an application server like Apache Tomcat allows Java-based web applications and servlets to run and serve dynamic content over HTTP.

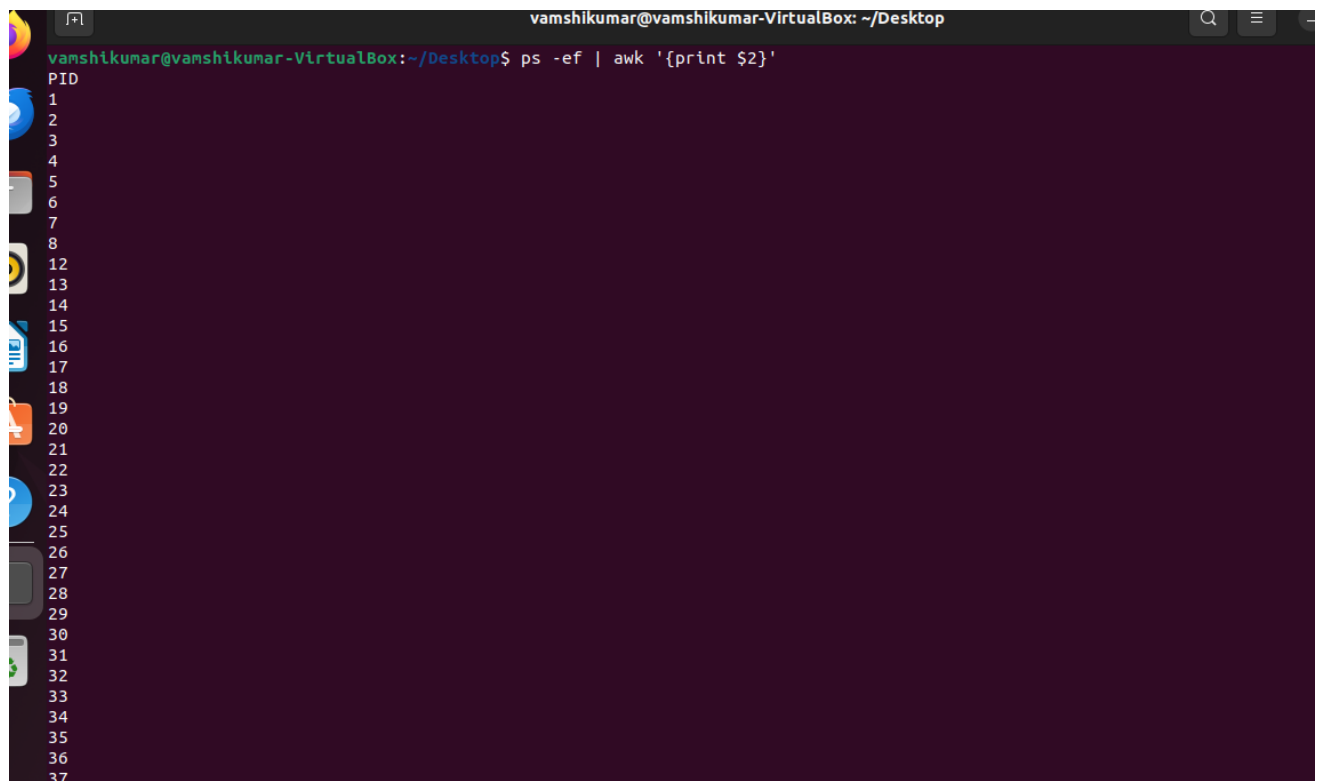
Command Execution and Verification

The command `sudo systemctl status tomcat9` checks the status of the Tomcat service, returning active (running) with process logs showing initializing ProtocolHandler ["http-nio-8080"] and Server startup in [2251] milliseconds. This verifies that Apache Tomcat is successfully initialized and running on port 8080.

Conclusion

Deploying an application server requires verifying the service state via `systemctl` to confirm that the Java runtime environment and web application directories are deployed and listening on the designated port.

30 .Print specific columns from a delimited file.



```
vamshikumar@vamshikumar-VirtualBox: ~/Desktop
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ps -ef | awk '{print $2}'
PID
1
2
3
4
5
6
7
8
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
```

Purpose

Printing specific fields or columns from a delimited file allows users to extract only relevant data fields from structured text files like CSVs.

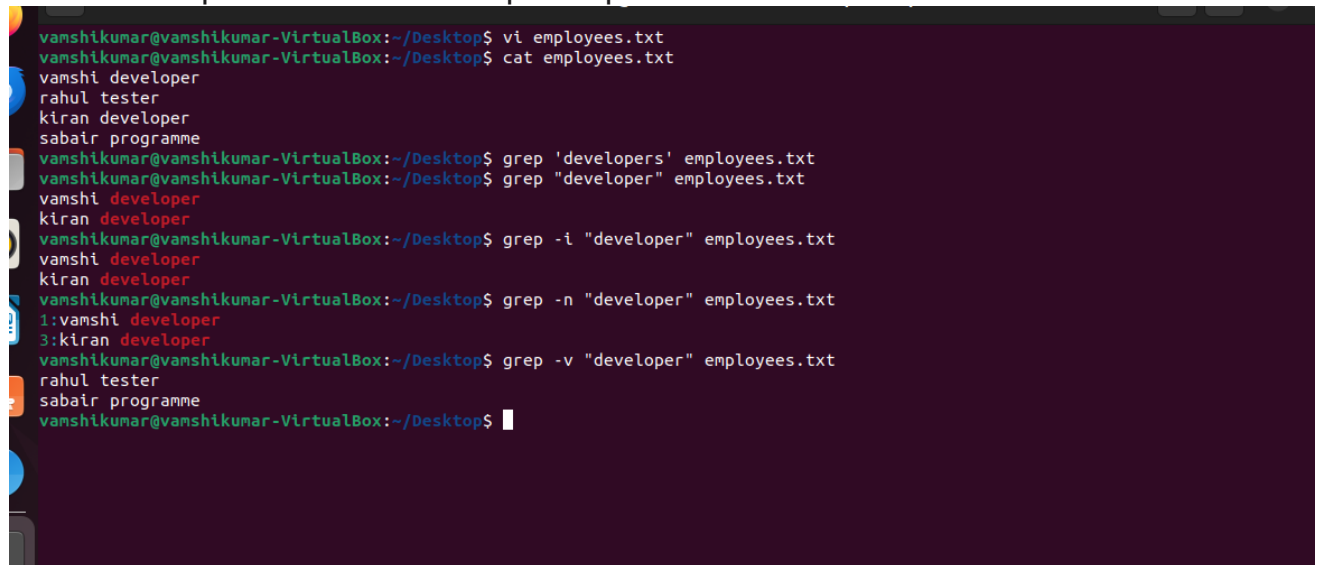
Command Execution and Verification

The command `cut -d',' -f1,3 f2` reads file `f2` using comma as the delimiter and extracts fields 1 and 3, outputting rows like `Cbash,58`, `bash,55`, and `kk,88`. This verifies that only the specified columns are selected and printed to stdout.

Conclusion

Using the `cut` command with `-d` to define the delimiter and `-f` to specify field numbers provides an efficient way to parse and extract targeted columns from structured tabular data.

31 .Filter and print lines based on a specific pattern or condition.



```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ vi employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
vamshi developer
rahul tester
kiran developer
sabair programme
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep 'developers' employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep "developer" employees.txt
vamshi developer
kiran developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -i "developer" employees.txt
vamshi developer
kiran developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -n "developer" employees.txt
1:vamshi developer
3:kiran developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -v "developer" employees.txt
rahul tester
sabair programme
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

Purpose

Filtering text using `grep` allows users to search for specific character patterns or keywords within a file and display matching or non-matching lines.

Command Execution and Verification

The command `grep "apple" test.text` searches for the exact string `apple`, while `grep -i "apple" test.text` performs a case-insensitive search to match terms like `Apple` or `APPLE`, both outputting matching lines such as `apple` and `apple juice`. Running `grep -v "apple" test.text` inverts the search to exclude lines containing `apple`, printing only `banana`, `orange`, and `grapes`.

Conclusion

The `grep` command provides versatile pattern matching, using flags like `-i` to ignore case sensitivity and `-v` to filter out unwanted lines from textual output.

32 Calculate and print the average, sum, or other statistics of a column.

```

vanshikumar@vanshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {sum += $2; count++} END {print sum , sum/count}'
172791 949.401
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {if ( $2 > max ) max = $2 } END {print max}'
3461
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ s

```

Purpose

The primary goal of using awk in this context is to perform statistical calculations, such as summing values and calculating the average across a specific column of data in a text file.

Command Execution and Verification

The command `awk '{sum += $2} END {print "sum =", sum}' marks.txt` iterates through each line of marks.txt, adds the values in column 2, and outputs `sum = 300` upon completion. To find the mean, `awk '{sum += $2; count++} END {print "Average =", sum/count}' marks.txt` tracks both total sum and line count, successfully outputting `Average = 60`.

Conclusion

Using awk with column variable referencing and an END block provides an efficient way to perform mathematical calculations and generate summary metrics directly from structured text data.

33 Perform string manipulation, such as extracting substrings or changing case.

```

vanshikumar@vanshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {sum += $2; count++} END {print sum , sum/count}'
172791 949.401
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {if ( $2 > max ) max = $2 } END {print max}'
3461
vanshikumar@vanshikumar-VirtualBox:~/Desktop$ s

```

Purpose

Performing string manipulation in Bash allows users to extract substrings, check variable lengths, and transform text cases using parameter expansion and stream editors like `tr` and `sed`.

Command Execution and Verification

Setting `name="chunduru"` and executing `echo "${name:0:4}"` extracts the first four characters, outputting `chun`, while `echo "${#name}"` measures and prints the string length of 8. Case conversion using `echo "$name" | tr '[:upper:]' '[:lower:]'` converts all characters to lowercase (`chunduru`), and running `echo "Hello Chunduru" | sed 's/chunduru/world/'` attempts pattern replacement but outputs `Hello Chunduru` unchanged due to case sensitivity.

Conclusion

Bash parameter expansion provides efficient built-in substring extraction and length calculation, while external tools like `tr` and `sed` extend text manipulation capabilities for case conversion and string substitution.

34 Count the occurrences of a specific pattern in a file.

```
developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -o "developer" employees.txt | wc employees.txt
 4  8 63 employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -ci "developer" employees.txt
2
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

Purpose

Counting occurrences of a specific pattern in a file allows users to determine how many lines or total individual instances contain a target keyword or phrase.

Command Execution and Verification

The command `grep -c "apple" test.text` counts matching lines and outputs 3, verifying that there are three lines containing the string "apple". Running `grep -oi "apple" test.text | wc -l` uses `-o` to extract every individual match on a separate line and `-i` to ignore case, piping the results to `wc -l` to verify a total count of 3 occurrences across the file.

Conclusion

Using `grep -c` provides a quick count of lines matching a pattern, while piping `grep -oi` into `wc -l` counts every individual instance regardless of line structure or letter casing.

35 Sort lines based on a specific field or column.

```

developer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -o "developer" employees.txt | wc employees.txt
 4  8 63 employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ grep -ci "developer" employees.txt
 2
vamshikumar@vamshikumar-VirtualBox:~/Desktop$

```

Purpose

Sorting data in structured files allows users to order records numerically or alphabetically based on a specific key or column value.

Command Execution and Verification

The command `sort -k2 -n marks.txt` sorts the lines in `marks.txt` numerically using the second column (`-k2 -n`), displaying records in ascending order from 60 to 90 (anil 60 to suresh 90). Running `sort -k2 -nr marks.txt` adds the `-r` flag to reverse the sort order, successfully outputting the records in descending order from 90 to 60 (suresh 90 to anil 60).

Conclusion

Using `sort` with the `-k` option targets specific columns for organization, while combining flags like `-n` for numerical sorting and `-r` for reverse ordering enables precise control over data sequence.

36 Merge multiple files based on a common field or column.

```

vamshikumar@vamshikumar-VirtualBox:~/Desktop$ vi employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ vi departments.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
101 Vamshi
102 Kiran
103 Rahul
104 Sabair
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat departments.txt
101 Developer
102 Analyst
103 Tester
104 Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sort -k1 employees.txt
101 Vamshi
102 Kiran
103 Rahul
104 Sabair
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sort -k1 departments.txt
101 Developer
102 Analyst
103 Tester
104 Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ join employees.txt departments.txt
101 Vamshi Developer
102 Kiran Analyst
103 Rahul Tester
104 Sabair Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$

```

Purpose

Merging files using the `join` command combines lines from two text files that share a common key or field value.

Command Execution and Verification

The command `join file1.txt file2.txt` initially fails because `join` requires input files to be sorted numerically or alphabetically beforehand. To fix this, running `sort -k1 -n file1.txt -o file1.txt` and `sort -k1 -n file2.txt -o file2.txt` sorts both files in place by their first field.

Re-running join file1.txt file2.txt successfully matches corresponding records based on column 1 and outputs the combined dataset.

Conclusion

The join command provides an efficient way to merge datasets on a shared key, provided both input files are pre-sorted along the join field using sort.

37 Substitute text in a file using search and replace.

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed 's/Developer/Engineer/' employees.txt
Vamshi is a Engineer
Rahul is a Tester
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
Vamshi is a Developer
Rahul is a Tester
Kiran is a Developer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed -i 's/Developer/Engineer/g' employees.txt
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
Vamshi is a Engineer
Rahul is a Tester
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

Purpose

Replacing or substituting text within a file allows users to swap specific words or phrases dynamically without manually editing the file line by line. Command Execution and Verification

The command `sed 's/ravi/kiran/g' test.txt` searches for all instances of "ravi" in test.txt and replaces them with "kiran", displaying the modified text in stdout while leaving the original file unchanged. Running `sed -i 's/ravi/kiran/g' test.txt` applies the -i flag to perform the substitution in-place directly on test.txt, which is verified by running `cat test.txt` to display updated lines containing "hello kiran" and "welcome kiran".

Conclusion

Using sed with the `s/pattern/replacement/g` syntax performs global search-and-replace transformations, while adding the -i option permanently saves those modifications directly to the source file.

38 Delete specific lines based on a pattern or line number.

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt
Vamshi is a Engineer
Rahul is a Tester
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed '2d' employees.txt
Vamshi is a Engineer
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed 's/Engineer/d' employees.txt
sed: -e expression #1, char 12: unterminated 's' command
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed '/Engineer/d' employees.txt
Rahul is a Tester
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

Purpose

Deleting specific lines in a file using sed allows users to remove unwanted content based on matching text patterns or exact line numbers.

Command Execution and Verification

The command `sed -i '/ravi/d' test.txt` deletes all lines containing the pattern "ravi", while `sed -i '2d' test.txt` removes line 2 specifically from the file. Running `sed -i '2,4d' test.txt` deletes a range of lines from line 2 through line 4 inclusive, which is verified using `cat test.txt` to display the remaining lines in the file.

Conclusion

Using sed with the d address command provides precise control for line removal, supporting pattern matching, single line deletion, and multi-line range removal directly within source files.

39Append or insert text before or after a specific pattern or line.

```
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sed -i '2a New Employee' employees.txt
Vamshi is a Engineer
New Employee
Rahul is a Tester
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sed '2a New Employee' employees.txt
Vamshi is a Engineer
Rahul is a Tester
New Employee
Kiran is a Engineer
Sabair is a Programmer
vamshikumar@vamshikumar-VirtualBox: ~/Desktop$ sed -i '/Rahul/i New Employee' employees.txt
```

Purpose

Appending or inserting text in a file using sed allows users to place new lines above or below specific matching patterns or specified line numbers.

Command Execution and Verification

The command `sed -i '/ravi/i Welcome' test.txt` inserts the text "Welcome" on a new line immediately before the line matching "ravi", while `sed -i '2a Hello' test.txt` appends "Hello" on a new line directly after line 2. Running `cat test.txt` verifies that the new lines "Welcome" and "Hello" are added at the precise target locations in the file.

Conclusion

Using sed with the `i` command inserts content before a target line, whereas using the `a` command appends content after a target line, offering flexible line-level text insertion.

40 Print only specific lines from a file.

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed -n '2p' employees.txt
Rahul is a Tester
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed -n '2,4p' employees.txt
Rahul is a Tester
New Employee
Kiran is a Engineer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed -n '/Engineer/p' employees.txt
Vanshi is a Engineer
Kiran is a Engineer
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

Purpose: To learn and demonstrate basic Linux text-processing commands like sed, head, tail, and cat.

Commands: `sed -n '2p' test.txt`, `sed -n '2,4d' test.txt`, `cat test.txt`, `head -n 1 test.txt`, `tail -n 1 test.txt`.

Conclusion: These commands help display, delete, and view specific lines or portions of a text file efficiently.

41 Copy file from linux to windows machine

```
vamshikumar@vamshikumar-VirtualBox: ~/Desktop
Building dependency tree... Done
Reading state information... Done
458 packages can be upgraded. Run 'apt list --upgradable' to see them.
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  openssh-client
1 upgraded, 4 newly installed, 0 to remove and 457 not upgraded.
Need to get 752 kB/1,656 kB of archives.
After this operation, 6,056 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-server amd64 1:8.9p1-3ubuntu0.16 [436 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term all 6.3-2ubuntu0.2 [267 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 ssh-import-id all 5.11-0ubuntu1 [10.1 kB]
Fetched 752 kB in 2s (329 kB/s)
Preconfiguring packages ...
(Reading database ... 204158 files and directories currently installed.)
Preparing to unpack .../openssh-client_1%3a8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-client (1:8.9p1-3ubuntu0.16) over (1:8.9p1-3ubuntu0.10) ...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1%3a8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-sftp-server (1:8.9p1-3ubuntu0.16) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a8.9p1-3ubuntu0.16_amd64.deb ...
Unpacking openssh-server (1:8.9p1-3ubuntu0.16) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.3-2ubuntu0.2_all.deb ...
Unpacking ncurses-term (6.3-2ubuntu0.2) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu1) ...
Setting up openssh-client (1:8.9p1-3ubuntu0.16) ...

vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-08-11 06:29:46 IST; 43s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 2976 (sshd)
      Tasks: 1 (limit: 2268)
     Memory: 1.7M
        CPU: 23ms
    CGroup: /system.slice/ssh.service
            └─2976 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 11 06:29:46 vamshikumar-VirtualBox systemd[1]: Starting OpenBSD Secure Shell server...
Aug 11 06:29:46 vamshikumar-VirtualBox sshd[2976]: Server listening on 0.0.0.0 port 22.
Aug 11 06:29:46 vamshikumar-VirtualBox sshd[2976]: Server listening on :: port 22.
Aug 11 06:29:46 vamshikumar-VirtualBox systemd[1]: Started OpenBSD Secure Shell server.
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ssh ss -lnp | grep :22
ssh: Could not resolve hostname ss: Temporary failure in name resolution
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sudo ss -lnp | grep :22
LISTEN 0      128          0.0.0.0:22      0.0.0.0:*       users:((("sshd",pid=2976,fd=3))

LISTEN 0      128          [::]:22        [::]:*          users:((("sshd",pid=2976,fd=4))

1738@IDEAPAD MINGW64 ~
$ cp -P 2222 vamshikumar@127.0.0.1:/home/vamshikumar/Desktop/test.txt ~/Downloads
vamshikumar@127.0.0.1's password:
test.txt
100% 25 2.4KB/s 00:00

1738@IDEAPAD MINGW64 ~
$ ls ~/Downloads/test.txt
/c:/Users/91738/Downloads/test.txt

1738@IDEAPAD MINGW64 ~
$ cd Downloads/

1738@IDEAPAD MINGW64 ~
$ ssh vamshikumar@10.0.2.15
ssh: connect to host 10.0.2.15 port 22: Connection timed out

1738@IDEAPAD MINGW64 ~
$ ^C

1738@IDEAPAD MINGW64 ~
$ ssh -p 2222 vamshikumar@127.0.0.1
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:MU5HGs7CUSL1rWpqqHizmDARGeo+8w26r4o5F4Q1xD8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
vamshikumar@127.0.0.1's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

464 updates can be applied immediately.
387 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```



```
awk: line 1: syntax error at or near '}'  
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {sum += $2; count++} END {print sum , sum/count}'  
172791 949.401  
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ ps -ef | awk 'NR>1 {if ( $2 > max ) max = $2 } END {print max}'  
3461  
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ S
```

```
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ cat employees.txt  
Vamshi is a Engineer  
Rahul is a Tester  
Kiran is a Engineer  
Sabair is a Programmer  
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed '2d' employees.txt  
Vamshi is a Engineer  
Kiran is a Engineer  
Sabair is a Programmer  
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed 's/Engineer/d' employees.txt  
sed: -e expression #1, char 12: unterminated `s' command  
vamshikumar@vamshikumar-VirtualBox:~/Desktop$ sed '/Engineer/d' employees.txt  
Rahul is a Tester  
Sabair is a Programmer  
vamshikumar@vamshikumar-VirtualBox:~/Desktop$
```

Purpose: To practice awk and sed commands for displaying, modifying, deleting, counting, and processing text in a file.

Commands: awk, sed -i 's/old/new/g', sed -i '2d', sed -i '/ravi/d', sed -i '2,4p', and sed -i '2i Hello'.

Conclusion: awk and sed are powerful Linux utilities used to efficiently search, edit, filter, and manipulate text files.